

Institute of Technology of Cambodia

Department of Foundation Years

Syllabus

Time Series Analysis and Forecasting

Academic 2026-2027

Course Specification on Time Series Analysis and Forecasting

Course Information

1. **Programme Tittle**	Engineering Degree in Data Science		
2. Course Title	Time Series Analysis and Forecasting		
3. **Course Code**	AMSI51TSA	4. **No. of Credits**	3(2, 0.5, 0.5)
5. Pre-requisites (If any)	Statistics and Probability		
6. Course Instructor (Name)	Dr. Sim Tepmony	**7. Qualification (last degree obtained)**	Doctoral Degree
8. **Email**	tepmony.sim@itc.edu.kh	9. **Telephone No.**	(+855)81988778
10. **Other Course Teacher(s)/Lecturer(s) (If any)**	Mr. Nhim Malai, Email: nhim.malai@itc.edu.kh, Tel: (+855)89820749 Dr. MOM Kannara, Email: kannara.mom@gmail.com, Tel: (+855)10838002		
11. **Course Type (optional)**	Major ☐ Major ☐ Core ☑ Elective ☐ Specialization ☐		
12. **Course Availability**	1st Semester ☑ 2nd Semester ☐ or Term		
13. **Course description/ Synopsis**	This course covers topics in the time series analysis and some statistical techniques on forecasting. These are time series regression, decomposition methods, exponential smoothing, and the Box-Jenkins forecasting methodology. The principal objective of the course is to introduce graduate and advanced undergraduate students in mathematics, economics, business, engineering, meteorology, various areas of social science, and any other field where analysis and research of time series are prevalent to many approaches of analyzing time series data. And at the same time, equip them with adequate statistical tools and knowledge to make precise decisions, based on forecasts obtained from the statistical analysis of historical data.		

14. Course Learning Outcomes

Here are the three CLOs of this course.

Description of the course learning outcomes – CLOs At the end of the course, students are able to:		Major Domaine	Learning Domaine
CLO1	Be able to analyse any time series data using various statistical approaches	MD1, MD2, and MD5	LD1, LD2 and LD10
CLO2	Be able to generate reasonable forecast values	MD1, MD3 and MD5	LD1, LD3, and LD10
CLO3	Be able to make concise decisions based on forecasts obtained	MD1, MD2, MD3 and MD5	LD1, LD2, LD3, and LD10

15. Mapping of the course learning outcomes to the programme Learning Outcome, Teaching Methods and Assessment methods

CLO	Programme Learning Outcome										Teaching Method	Assessment methods
	PLO1	PLO2	PLO3	PLO 4	PLO 5	PLO 6	PLO7	PLO8	PLO9	PLO1 0		
CLO1	✓	✓								✓	Lecture TD/TP Group work	Quizzes/Midterm Whiteboard Presentation Final
CLO2	✓		✓							✓	Lecture TD/TP Group work	Quizzes/Midterm Whiteboard Presentation Final

CLO3	✓	✓	✓							✓	Lecture TD/TP Group work	Quiz/Final exam Whiteboard Presentation Final

16. Distribution of Student Learning Time (SLT)

Course Content Outline and Subtopics		CLO*	Learning and Teaching Activities							Total SLT
			Physical Class			Online/Technology-mediated (Synchronous) (if applicable)			NF2F Independent Learning	
			L	**T**	**P**	**L**	**T**	**P**		
1	Lecture 1: Introduction		**2**	**1**	**1**				**2**	**6**
2	Lecture 2: Fundamental Statistical Concepts		**2**	**1**	**1**				**2**	**6**
3	Lecture 3: Trend Analysis		**2**	**1**	**1**				**2**	**6**
4	Lecture 4: Model for Non-Stationary Time Series		**2**	**1**	**1**				**2**	**6**
5	Lecture 5: Model Specification or Identification		**2**	**1**	**1**				**2**	**6**
6	Lecture 6: Model Specification		**4**	**2**	**2**				**2**	**10**
7	Lecture 7: Parameter Estimation		**2**	**1**	**1**				**2**	**6**

8	Lecture 8: Model Diagnostics		**2**	**1**	**1**				**2**	**6**
9	Lecture 9: Forecasting		**4**	**2**	**2**				**2**	**10**
10	Lecture 10: ARIMA Models		**2**	**1**	**1**				**2**	**6**
11	Lecture 11: Seasonal Models		**4**	**2**	**2**				**2**	**10**
12	Lecture 12: Time Series Regression Models		**2**	**1**	**1**				**2**	**6**
13	Lecture 13: Deep learning with Time Series Analysis		**2**	**1**	**1**				**2**	**6**
									Sub Total SLT:	**90**

Continuous Assessment		%	Face-to-Face (F2F)		NF2F Independent Learning for Assessment	
			Physical Class	Online/Technology-mediated (Synchronous) (if applicable)		
1	Quiz and Class Activity	**20%**	**2**		**4**	**6**
2	Midterm	**20%**	**2**			**2**
3	Project (Report and Presentation)	**30%**	**2**		**16**	**18**
					Sub Total SLT:	**26**

Final Assessment	%	Face-to-Face (F2F)			
		Physical Class	Online/Technology-mediated (Synchronous) (if applicable)	NF2F Independent Learning	

2	Final	**30%**	**2**			**2**
					Sub Total SLT:	**2**
					Grand Total SLT:	118
					Grand Total SLT/40	2.95

Please check (✓) if this course is Industrial/Clinical Placement/ Practicum using 50% of effective Learning Time(ELT)

17. **Course Assessment Plan**

Course: Time Series Analysis and Forecasting, Year 5, Semester 1

CLOs	CLOs statement	PLOs	Assess. Methods	Weightage (%)	T&L* Strategy/ Technique	Level of Learning (C/A/P*)	Topics	SLT** (h)
1	Be able to analyse any time series data using various statistical approaches	1, 2, 10	Attendance and Quiz	3%	• Lecture • Think-Pair-Share • Peer teaching	(3/0/2)	1, 2	16
			Class Activities	6%			1, 2	
			Project Report and Presentation	6%			1, 2	
			Midterm	10%			1, 2	
2	Be able to generate reasonable forecast values	1, 3, 10	Attendance and Quiz	3%	• Lecture • Think-Pair-Share • Peer teaching	(2/0/3)	1-13	108
			Class Activities	6%			1-13	
			Project Report and Presentation	6%			1-13	
			Midterm	10%			1-13	

3	Be able to make concise decisions based on forecasts obtained	1, 2, 3, 10	Attendance and Quiz	4%	• Lecture • Think-Pair-Share • Peer teaching	(2,0,0)	3-13	88
			Class Activities	8%			3-13	
			Project Report and Presentation	8%			3-13	
			Fial	30%			3-13	

*T&L: Teaching and Learning

**SLT: Student Learning Time

18.Course Assessment Plan

Weeks	Sessions	CLOS	LLOs	Content	Teaching Activities	Learning Activities	Assessment	Materials
1	1-4	CLO1 CLO2	• Ability to understand concept of time series, various time series model • Ability to use a model building strategy • Ability to illustrate the time series plots in history	Lecture 1: Introduction	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
2	5-8	CLO1 CLO2	• Ability to understand Fundamental	Lecture 2: Fundamental	• Lecture presentation	• Listen and note taking	• Attendance • Class activities	• Slide presentation

			Time Series and Stochastic Processes. • Ability to explain Means, Variances, and Covariances. • Ability to understand Stationarity.	Statistical Concepts	• Tutorial practice • Class facilitating • Case study • Q&A	• Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Formative assessment • Quiz • Q&A	• LCD projector • Laptop • Slide pointer • Writing board
3	9-12	CLO2 CLO3	• Ability to understand Deterministic VS Stochastic Trends. • Ability to explain the Estimation of a Constant Mean. • Ability to understand Regression Methods. • Ability to Interpret Regression Output.	Lecture 3: Trend Analysis	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
4	13-16	CLO2 CLO3	• Ability to explain Stationary Time Series such as Weakly Stationary. • Ability to understand Moving Average Processes,	Lecture 4: Models for Stationary Time Series	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board

			Autoregressive Processes, and The Mixed Autoregression Moving Average Models. • Ability to explain the Invertibility of Moving Average Model.			• Q&A		
5	17-20	CLO2 CLO3	• Ability to understand the Stationarity Through Differencing. • Ability to illustrate the concept of ARIMA Models. • Ability to solve the problem related to Constant Terms in ARIMA Models.	Lecture 5: Model for Non-Stationary Time Series	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
6	21-24	CLO2 ClO3	• Ability to understand the Properties of the Sample Autocorrelation Function. • Ability to explain Partial and Extended Autocorrelation Functions.	Lecture 6: Model Specification or Identification	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board

			• Ability to explain Specification of Some Simulated Time Series. • Ability to understand Nonstationary and other Specification Models.					
7	25-28	CLO2 CLO3	• Ability to understand The Method of Moments, Least Squares Estimation. • Ability to Solve Problem related to Maximum Likelihood and Unconditional Least Squares. • Ability to explain the Properties of Estimates.	Lecture 7: Parameter Estimation	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
8	29-32		**Midterm Exam (Written Test)**					
9	33-36	CLO2 CLO3	• Ability to understand Residual Analysis. • Ability to explain the Overfitting and Parameter Redundancy.	Lecture 8: Model Diagnostics	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board

			• Ability to solve and illustrate the related problem with R.			• Report/Project presentation • Q&A		
10	37-40	CLO2 CLO3	• Ability to understand the Minimum Mean Square Error Forecasting • Ability to illustrate Deterministic Trends. • Ability to explain ARIMA Forecasting. • Ability to understand the Prediction Limit and Forecasting Illustrations. • Ability to solve Forecast Weights and Exponentially Weighted Moving Averages.	Lecture 9: ARIMA Models	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
11	41-44	CLO2 CLO3	• Ability to understand and explain each of ARIMA models. • Ability to understand Spurious Correction,	Lecture 10: Seasonal Models	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board

			Prewhitening and Stochastic Regression.			• Report/Project presentation • Q&A		
12	49-52	CLO2 CLO3	• Ability to explain the Seasonal ARIMA Models. • Ability to explain Multiplicative Seasonal ARMR Models. • Ability to solve problems related to Nonstationary Seasonal ARIMA Models. • Ability to illustrate the concept of Model Specification, Fitting and Checking.	Lecture 11: Time Seasonal Models	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
13	49-52	CLO2 CLO3	• Ability to understand Some Common Feature of Financial Time Series. • Ability to understand The ARCH(1) Model, GARCH Model. • Ability to explain the Maximum	Lecture 12: Time Series Regression Models	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board

			Likelihood Estimation. • Ability to illustrate Model Diagnostics.					
14	53-56	CLO2 CLO3	• Ability to explain recurrent neural network (RNN) and long short-term memory (LSTM) architecture • Ability to use Python to implement deep learning models for time-series forecasting	Lecture 13: Deep learning with time series analysis	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
15	57-60	CLO2 CLO3	• Ability to solve real world problems as a group with time series analysis • Ability to understand group work • Ability to explain/present group work/project	Project: Final Presentation	• Lecture presentation • Tutorial practice • Class facilitating • Case study • Q&A	• Listen and note taking • Individual homework • Group discussion and presentation • Report/Project presentation • Q&A	• Attendance • Class activities • Formative assessment • Quiz • Q&A	• Slide presentation • LCD projector • Laptop • Slide pointer • Writing board
16			**Prepare for Final Exam**					
17			**Final Exam**					

19.Require resources to deliver the course (E.g: QGIS sofware,AutoCAD, Python, Weather Station, Computer lab, etc.)

- Moodle

- Computer lab
- R/Python

- **Required References/Textbooks and Other Teaching and Learning materials**

20. Required Textbooks

Title	Author (s)	Publisher	Year	ISBN
Time series regression models: time series with application in R	Jonathan D. Cryer, Kung-Sik Chan	Springer	2008	978-0387759586
Time Series Analysis and Its Applications: With R Examples	Lay, D. C., Lay, S. R., McDonald, J. J.	Springer	2017	978-3319524511
The Analysis of Time Series: An Introduction with R	Chris Chatfield, Haipeng Xing	Chapman and Hall/CRC	2019	978-1498795630

Recommended references books/ Readings

Title	Author (s)	Publisher	Year	ISBN
An Introduction to Time Series Analysis and Forecasting: With Applications of SAS and SPPS	Robert Alan Yaffee, Monnie McGee	Academic Press	2000	978-0127678702

Forecasting, Time Series, and Regression (with CD-ROM)	Bruce L. Bowerman, Richard O'Connell, Anne Koehler	Thomson Brooks/Cole, Belmont, CA	2005	978-0534409777

21. Student responsibility

- Join the physical class or online class.
- Review video lectures, read more, do quiz in the Moodle.
- Prepare presentation for specific given TP (in group of 5 students)
- Review Lab. instruction before class

22.Rubric and Rating scale (iRubric)

CLO1 Evaluation Rubric		
Evaluation Categories (Assessment Method)	Evaluation Definition (Assessment Task)	Mark Allocation (Scoring)
Excellent (4)	Demonstrates mastery in identifying and applying multiple advanced statistical methods (e.g., decomposition, ARIMA, exponential smoothing). Provides clear justification for method choices and insights are well-supported by analysis.	21-25
Good (3)	Applies appropriate statistical methods and provides reasonable justification. Demonstrates good understanding, though analysis might lack depth or miss some advanced techniques.	16-20
Satisfactory (2)	Applies basic statistical methods (e.g., trend/seasonality identification, simple smoothing). Some errors or misunderstandings may exist. Limited justification or interpretation.	11-15
Needs Improvement (1)	Fails to apply appropriate statistical techniques. Analysis is incomplete or incorrect. Lacks understanding of fundamental time series components.	0-10

CLO2 Evaluation Rubric		
Evaluation Categories (Assessment Method)	Evaluation Definition (Assessment Task)	Mark Allocation (Scoring)
Excellent (4)	Produces accurate forecasts using appropriate models, with validation (e.g., cross-validation, AIC/BIC, residual diagnostics). Forecasts are realistic and well-justified.	21-25
Good (3)	Forecasts are mostly accurate and based on suitable models. Some validation is performed, but may not be comprehensive.	16-20
Satisfactory (2)	Forecasts are generated using basic models (e.g., naïve, moving average), but lack validation or detailed reasoning. May include minor errors.	11-15
Needs Improvement (1)	Forecasting methods are inappropriate, overly simplistic, or incorrectly applied. Results are unreliable or unjustified.	0-10

CLO3 Evaluation Rubric		
Evaluation Categories (Assessment Method)	Evaluation Definition (Assessment Task)	Mark Allocation (Scoring)
Excellent (4)	Makes clear, data-driven decisions aligned with forecast results. Demonstrates strong critical thinking and ability to communicate implications effectively.	41-50
Good (3)	Decisions are reasonable and based on forecasts, though some insights may be superficial or lack clarity.	31-40
Satisfactory (2)	Decisions are somewhat related to forecasts but may be vague, poorly justified, or contain interpretive errors.	21-30
Needs Improvement (1)	Decisions are not clearly linked to forecast outcomes. Shows little understanding of how to interpret or use forecast information.	0-20

22. Course policy

- **Attendance**: Attending on-site classes is strongly encouraged, and each week students must watch assigned videos and read the assigned chapter, and students need to fill in the preparation outline before class. Students will be required to do discussion, sharing solutions, formative assessments.
- **Academic Integrity & Collaboration**: Students must do problem sets, final project and quizzes (exams) only by himself/herself. Questions might be asked by the instructor during/after the submission of the solutions, final project, problem sets etc. Students shall be guilty of violating the honor code if they: (1) Present the work of others as their own, (2) Use or obtain unauthorized assistance in any academic work, and (3) Give unauthorized assistance to other students.
- **Lab & Homework**: Student needs to submit problem sets on time and read the assigned reading homework before class. Late homework submission will result in getting zero score. You are encouraged to discuss the course material with other students. However, you are required to work independently to solve the assigned problem for lab and homework. Copying or plagiarism will be treated as academic dishonesty and will be handled according to institution regulations.
- **Exam**: You are encouraged to discuss the course material with other students and lecturers before the exam. However, you are required to work independently on the midterm and final exams. Students are not allowed to use phones, computers, or book during final exams.
- **Final project**: Students are encouraged to have discussions with lecturers and classmates about the problems. However, they need to formulate their own solution to solve final project.
- **Penalty**: The penalty for violating the institute honor code is severe. Any student violating the honor code is subject to receive a failing grade for the course and will be reported to The Office of Student Affairs. If you are unclear about whether a particular situation may constitute an honor code violation, you should ask the instructor.

25. Date: 06/October/2026